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Inverting the Formalization Workflow

Prototyping an MPC Protocol Idea in Rocq with an LLM Agent

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THE CORE IDEA

Formalization doesn't have to be last.

TRADITIONAL



INVERTED WORKFLOW



THE COLLABORATION

What each provides



HUMAN

Judgment, architecture,
what to axiomatize



AI AGENT

Breadth across unfamiliar
mathematical domains



PROOF ASSISTANT

Guarantees correctness
up to chosen axioms